8.20 Solution: The semicircular loop can be parameterized by the angle measured from the short side, and the points on the circle and the currents in the circle are given by

$$\mathbf{x} = (R\sin\phi, h + R\cos\phi), \quad \mathbf{J} = I_0(\cos\phi, -\sin\phi).$$

(a) The normalized fields for TM modes are (we are only interested in x and y components), by Eq. (8.135),

$$E_{xmn} = \frac{2\pi m}{\gamma_{mn} a\sqrt{ab}} \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right), \quad E_{ymn} = \frac{2\pi n}{\gamma_{mn} b\sqrt{ab}} \sin\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right).$$

We can find the amplitudes as, Eq. (8.146),

$$\begin{aligned}
A_{mn} &= -\frac{Z_{\mathrm{TM}}}{2}\int \mathbf{J}\cdot\mathbf{E}d^3x \\
&= -\frac{Z_{\mathrm{TM}} I_0 R}{2\gamma_{mn}\sqrt{ab}}\int_0^\pi \left[\frac{m}{a}\cos\phi\cos\left(\frac{m\pi R}{a}\sin\phi\right)\sin\left(\frac{n\pi}{b}(h+R\cos\phi)\right)\right. \\
&\qquad\qquad \left. -\frac{n}{b}\sin\phi\sin\left(\frac{m\pi R}{a}\sin\phi\right)\cos\left(\frac{n\pi}{b}(h+R\cos\phi)\right)\right]d\phi \\
&= -\frac{Z_{\mathrm{TM}} I_0 R}{2\gamma_{mn}\sqrt{ab}}\int_0^\pi \left[\frac{m}{a}\cos\phi\cos\left(\frac{m\pi R}{a}\sin\phi\right)\sin\left(\frac{n\pi h}{b}\right)\cos\left(\frac{n\pi R}{b}\cos\phi\right)\right. \\
&\qquad\qquad +\frac{m}{a}\cos\phi\cos\left(\frac{m\pi R}{a}\sin\phi\right)\cos\left(\frac{n\pi h}{b}\right)\sin\left(\frac{n\pi R}{b}\cos\phi\right) \\
&\qquad\qquad -\frac{n}{b}\sin\phi\sin\left(\frac{m\pi R}{a}\sin\phi\right)\cos\left(\frac{n\pi h}{b}\right)\cos\left(\frac{n\pi R}{b}\cos\phi\right) \\
&\qquad\qquad \left. +\frac{n}{b}\sin\phi\sin\left(\frac{m\pi R}{a}\sin\phi\right)\sin\left(\frac{n\pi h}{b}\right)\sin\left(\frac{n\pi R}{b}\cos\phi\right)\right]d\phi.
\end{aligned}$$

It can be seen that the first and last two integrands are both odd with respect to $\phi = \pi/2$ and therefore their integrals will vanish. For the second and third integrands, notice that

$$\frac{m}{a}\cos\phi\cos\left(\frac{m\pi R}{a}\sin\phi\right)\sin\left(\frac{n\pi R}{b}\cos\phi\right) = \frac{1}{\pi R}\sin\left(\frac{n\pi R}{b}\cos\phi\right)d\left[\sin\left(\frac{m\pi R}{a}\sin\phi\right)\right],$$

and

$$-\frac{n}{b}\sin\phi\sin\left(\frac{m\pi R}{a}\sin\phi\right)\cos\left(\frac{n\pi R}{b}\cos\phi\right) = \frac{1}{\pi R}\sin\left(\frac{m\pi R}{a}\sin\phi\right)d\left[\sin\left(\frac{n\pi R}{b}\cos\phi\right)\right].$$

Then, these two integrands can be combined to form

$$\frac{1}{\pi R}\cos\left(\frac{n\pi h}{b}\right)d\left[\sin\left(\frac{m\pi R}{a}\sin\phi\right)\sin\left(\frac{n\pi R}{b}\cos\phi\right)\right],$$

which upon integration from 0 to π will give an integral of 0.

There is an intuitive explanation to this result. Consider the integral of to determine the expansion coefficients, which can be reduced to a line integral along the semicircular loop. Applying the Stokes' theorem and the Faraday's law,

$$\int \mathbf{J}\cdot\mathbf{E}d^3x = \int \mathbf{E}\cdot d\mathbf{l} = \int (\nabla\times\mathbf{E})\cdot\mathbf{n}da = -\frac{\partial}{\partial t}\int \mathbf{B}\cdot\mathbf{n}da,$$

where $\mathbf{n}$ is in $\pm z$ direction. In TM modes, there is no longitudinal magnetic field, *i.e.*, $B_z = 0$. Therefore, the expansion coefficients must all vanish and there is no excitation of TM modes.

(b) Since $a > b$, the lowest TE mode is the TE_{10} mode, where the only nonvanishing electric field is

$$E_{y10} = \frac{\sqrt{2}\pi}{\gamma_{10} a\sqrt{ab}} \sin\left(\frac{\pi x}{a}\right) = \sqrt{\frac{2}{ab}} \sin\left(\frac{\pi x}{a}\right).$$

The amplitude for this mode is

$$A_{10} = \frac{Z_{\mathrm{TE}} I_0 R}{\sqrt{2ab}} \int_0^{\pi} \sin\phi \sin\left(\frac{\pi R}{a}\cos\phi\right) d\phi = \frac{Z_{\mathrm{TE}} \pi I_0 R}{\sqrt{2ab}} J_1\left(\frac{\pi R}{a}\right),$$

which is independent of h.

(c) The transmitted power is

$$P = \frac{A_{10}^2}{2Z_{\mathrm{TE}}} = Z_{\mathrm{TE}} \frac{\pi^2 I_0^2 R^2}{2ab} J_1\left(\frac{\pi R}{a}\right)^2.$$

Since

$$J_1(x) \approx \frac{x}{2},$$

for $x \ll 1$, in our case, when $R \ll a$, we will have

$$P = Z_{\mathrm{TE}} \frac{\pi^2 I_0^2 R^2}{2ab} \left(\frac{\pi R}{2a}\right)^2 = \frac{I_0^2}{16} Z_{\mathrm{TE}} \frac{a}{b} \left(\frac{\pi R}{a}\right)^4.$$